Krylov subspace methods are the basis for iterative methods for eigenvalue problems, and also for solving linear systems.

Definition (Krylov subspace)**.** Given a matrix $A \in \mathbb{C}^{n\times n}$ and a starting vector $b \in \mathbb{C}^n$, the mth Krylov subspace is

$$\mathcal{K}_m(A, b) = \operatorname{span}\{b, Ab, A^2 b, \dots, A^{m-1}b\}.$$

An important advantage of Krylov methods is that they do not deal directly with A. Instead, they only require matrix–vector products involving A. This is especially helpful when A is large, since matrix–vector multiplication can be relatively cheap compared with forming or factorizing A.

1 Hessenberg Form

Definition (Hessenberg matrix)**.** A matrix $A = (a_{ij})$ is called *upper Hessenberg* if

$$a_{ij} = 0 \qquad \text{for all } i > j + 1.$$

It is called *lower Hessenberg* if

$$a_{ij} = 0 \qquad \text{for all } j > i + 1.$$

The Arnoldi iteration is a Krylov subspace iterative method that reduces A to upper Hessenberg form.

2 Arnoldi Iteration

For $A \in \mathbb{C}^{n\times n}$, a full Hessenberg factorization has the form

$$A = QHQ^*,$$

where H is upper Hessenberg and Q is unitary, i.e., $QQ^* = I$.

When n is large, we usually do not compute the full factorization. Instead, we consider only the first $m \ll n$ columns of the factorization

$$AQ = QH.$$

On the left-hand side, this only requires the matrix

$$Q_m \in \mathbb{C}^{n\times m}, \qquad Q_m = \begin{bmatrix} q_1 & q_2 & \cdots & q_m \end{bmatrix}.$$

On the right-hand side, the upper Hessenberg structure implies that we only need the first m columns of H. More precisely, we use the $(m + 1) \times m$ upper-left section

$$\widetilde{H}_m = \begin{bmatrix} h_{11} & h_{12} & \cdots & h_{1(m-1)} & h_{1m} \\ h_{21} & h_{22} & \cdots & h_{2(m-1)} & h_{2m} \\ 0 & h_{32} & \cdots & h_{3(m-1)} & h_{3m} \\ \vdots & \ddots & \ddots & \vdots & \vdots \\ 0 & \cdots & 0 & h_{m(m-1)} & h_{mm} \\ 0 & \cdots & \cdots & 0 & h_{(m+1)m} \end{bmatrix}.$$

Since $\widetilde{H}_m$ only interacts with the first $m+1$ columns of Q, we have

$$\boxed{AQ_m = Q_{m+1}\widetilde{H}_m.}$$

The mth column can be written as

$$Aq_m = h_{1m}q_1 + h_{2m}q_2 + \cdots + h_{mm}q_m + h_{(m+1)m}q_{m+1}.$$

Equivalently, assuming $h_{(m+1)m} \neq 0$,

$$q_{m+1} = \frac{Aq_m - h_{1m}q_1 - h_{2m}q_2 - \cdots - h_{mm}q_m}{h_{(m+1)m}}.$$

Remark. Arnoldi iteration is the Gram–Schmidt method applied to the Krylov sequence. It constructs the coefficients h_{ij} and the orthonormal vectors q_j.

3 The Algorithm

Arnoldi iteration

1: choose b arbitrarily, then $q_1 = b/\|b\|_2$
2: **for** $m = 1, 2, 3, \ldots$ **do**
3:     $v = Aq_m$
4:     **for** $j = 1, 2, \ldots, m$ **do**
5:         $h_{jm} = q_j^* v$
6:         $v = v - h_{jm}q_j$
7:     **end for**
8:     $h_{(m+1)m} = \|v\|_2$
9:     $q_{m+1} = v/h_{(m+1)m}$
10: **end for**

"This is akin to the modified Gram–Schmidt method because the updated vector v is used in line 5, rather than the raw vector Aq_m.

Also, in each iteration we only need to evaluate Aq_m and perform vector operations."

4 Ritz Values from Arnoldi Iteration

Question. How do we find eigenvalues from the Arnoldi iteration?

Answer. Let

$$H_m = Q_m^* A Q_m$$

be the $m \times m$ matrix obtained by removing the last row from $\widetilde{H}_m$. At each step m, we compute the eigenvalues of the Hessenberg matrix H_m. This is how `eigs` in Python/MATLAB works.

The eigenvalues of H_m are estimates for eigenvalues of A and are called *Ritz values*. The corresponding approximate eigenvectors are called *Ritz vectors*; if $H_m y = \theta y$, then the corresponding Ritz vector for A is $Q_m y$.

Just as with the power method, Ritz values typically converge to extreme eigenvalues of the spectrum.

5 Why Ritz Values Approximate Eigenvalues

We now examine why eigenvalues of H_m approximate extreme eigenvalues of A. Let $\mathcal{P}^m_{\text{monic}}$ denote the set of monic polynomials of degree m.

Theorem (Arnoldi approximation property)**.** The characteristic polynomial of H_m is the unique solution of the approximation problem

$$p^* = \arg\min_{p \in \mathcal{P}^m_{\text{monic}}} \|p(A)b\|_2.$$

Equivalently, the characteristic polynomial of H_m is the monic polynomial of degree m for which $\|p(A)b\|_2$ is minimal.

This theorem implies that the Ritz values, i.e., the eigenvalues of H_m, are the roots of the optimal polynomial

$$p^* = \arg\min_{p \in \mathcal{P}^m_{\text{monic}}} \|p(A)b\|_2.$$

Now consider what p^* should look like in order to minimize $\|p(A)b\|_2$.

A simple case

Suppose that A has only $m \ll n$ distinct eigenvalues and that

$$b = \sum_{i=1}^{m} \alpha_i v_i,$$

where v_i is an eigenvector corresponding to λ_i.

For $p \in \mathcal{P}^m_{\text{monic}}$, write

$$p(x) = c_0 + c_1 x + c_2 x^2 + \cdots + x^m$$

for some coefficients $c_0, c_1, \ldots, c_{m-1}$.

Applying this polynomial to the matrix A gives

$$\begin{aligned}
p(A)b &= \left(c_0 I + c_1 A + c_2 A^2 + \cdots + A^m\right) b \\
&= \sum_{i=1}^{m} \left(c_0 I + c_1 A + c_2 A^2 + \cdots + A^m\right) \alpha_i v_i \\
&= \sum_{i=1}^{m} \left(c_0 + c_1 \lambda_i + c_2 \lambda_i^2 + \cdots + \lambda_i^m\right) \alpha_i v_i \\
&= \sum_{i=1}^{m} \alpha_i p(\lambda_i) v_i.
\end{aligned}$$

Therefore, the polynomial $p^* \in \mathcal{P}_{\text{monic}}^m$ with roots at $\lambda_1, \lambda_2, \ldots, \lambda_m$ minimizes $\|p(A)b\|_2$, since

$$\|p^*(A)b\|_2 = 0.$$

This means that the Ritz values after m iterations are exactly the m distinct eigenvalues of A in this simple case.